KCE PATHWAYS · GRADE 3 · UNIT 5: DATA AND DESIGN · WEEK 24 · MATH

Barbecue Festival in Kansas City — Teacher Lesson Plan & Answer Key

Learning objectives: Students solve two-step word problems using addition, subtraction, multiplication and division, write and solve equations with a letter standing for the unknown number (doing the × or ÷ part first), and use rounding to the nearest ten or hundred to estimate and decide whether an answer is reasonable.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Title and name: the student types first and last name so results go to KingCretot Experience. Read the Pathway Points rules together: 2 tries per question, points only for a correct first try, and the answer and reason appear after a second miss. Introduce the barbecue festival in Kansas City, Missouri, with pitmaster Mr. Hargrove and his helpers Gemma, Hector and Hattie.
2. Warm-Up: Shapes and Equal Parts Review (10 pts, Week 23 skill): naming quadrilaterals and unit fractions. If a student misses both, review "rectangle: 4 right angles; rhombus: 4 equal sides; square: both."
3. Picture It: Barbecue Festival in Kansas City. The helpers count plates, share food into equal trays, find unknown numbers and estimate to check their work. Ask: what is the hidden question in a two-step problem?
4. Word Wrangler: 6 vocab cards (two-step problem, equation, unknown, round, estimate, reasonable).
5. Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade3/prepost/KCE_Math_PrePost_G3W24.pdf) and completes Page 1 (Pre) before any teaching. Do not help with answers yet; it shows what they already know.
6. Two Steps with Adding and Subtracting (10 pts): using a letter for the unknown (250 − 85 − 97 = r, so r = 68). Tip: find the hidden question first, then use that answer.
7. Two Steps with Multiplying and Dividing (10 pts): equal groups, then add or subtract (6 × 8 + 15 = 63). Tip: when an equation has × or ÷ and also + or −, do the × or ÷ part first.
8. Writing the Equation (10 pts): choosing the equation that matches a story and solving for the letter, which can be at the start, middle or end.
9. Is My Answer Reasonable? Estimate by Rounding (10 pts): rounding to the nearest ten or hundred and comparing the exact answer to the estimate. Tip: look at the digit to the right; 5 or more rounds up.
10. Pit Challenge: Two-Step Problems (10 pts): mixed operations, money and change. Ask the student to estimate first and check that the answer is close.
11. Bonus Practice: The Sauce Table (XP only, not scored).
12. Match-Up Round (10 pts, 2 per first-try match): Set 1 matches vocabulary to meanings, Set 2 matches numbers to their rounded values, Set 3 matches equations to the value of n.
13. Fix the Sentence (10 pts, 5 each): typed answers are checked exactly (capital letters for names of people and places and the title Mr., end mark, list commas, and one verb fix); curly quotes and extra spaces are forgiven. Both serial-comma and no-serial-comma list forms are accepted. Remind the student to keep the same words in the same order.

14. Speed Arena and Keyboard Corral (XP only, not scored).
15. Boss Battle, Part 1: The Dinner Rush (10 pts, 5 each): two-step problems, solving an equation for t, and judging or fixing an estimate (288 rounds to 300, not 200).
16. Boss Battle, Part 2: Closing Time (10 pts, 5 each): two-step problems and find-the-mistake items where the student types the correct answer (for example, 5 × 6 + 4 = 34, not 50).
17. Pencil Break (Post): the student turns to Page 2 (Post) of the same sheet (grade3/prepost/KCE_Math_PrePost_G3W24.pdf) and completes it, plus the ★ Challenge if time allows, before seeing the score. The sheet buttons let the student open or print it again.
18. Results (Festival Day Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows "Festival Hero!" with the congratulations message "Festival hero! You solved every problem at the barbecue stand!" and a big confetti burst; below 85 shows "Great Effort Today!" with encouragement ("Every pitmaster gets better with practice."), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: SHAPES AND EQUAL PARTS REVIEW (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: The sign for Mr. Hargrove's stand is a quadrilateral with 4 right angles. Its sides are 5 feet, 2 feet, 5 feet, and 2 feet long. Type the best name for its shape.	**rectangle** (also accepted: a rectangle, an rectangle, it is a rectangle, it's a rectangle, it is an rectangle)	It has 4 right angles, so it is a rectangle. Its sides are not all the same length, so it is not a square or a rhombus.
w2_s0	Set 1: Which explanation is correct? Hattie says a square napkin is not a rhombus. [Hattie is wrong. A square has 4 equal sides, so it is also a rhombus. / Hattie is right. A rhombus can never have right angles. / Hattie is right, because a square is a rectangle and cannot be anything else. / Hattie is wrong, because a square has 3 sides.]	**Hattie is wrong. A square has 4 equal sides, so it is also a rhombus.**	A rhombus is any quadrilateral with 4 equal sides. A square has 4 equal sides, so it is a rhombus too.
w1_s1	Set 2: A square pan of cornbread is cut into 6 equal pieces. What fraction of the pan is one piece?	**1/6** (also accepted: 1 / 6, 1 sixth, one sixth, one-sixth, a sixth)	6 equal pieces make the whole pan, so each piece is 1/6 of the pan.
w2_s1	Set 2: Find the mistake. Hector says a paper place mat with 4 equal sides and no right angles is a square. What went wrong? [A square needs 4 right angles too. A shape with 4 equal sides and no right angles is a rhombus. / Nothing is wrong. Any shape with 4 equal sides is a square. / It is a triangle, because it has no right angles. / It is a rectangle, because it has 4 sides.]	**A square needs 4 right angles too. A shape with 4 equal sides and no right angles is a rhombus.**	Equal sides make a rhombus. It is a square only if it also has 4 right angles.
w1_s2	Set 3: Two steps! Gemma stacks 3 rectangle-shaped serving trays and 2 square trays. How many right angles do all the trays have in all?	**20** (also accepted: 20 right angles, 20 angles)	Rectangles and squares each have 4 right angles. 3 + 2 = 5 trays, and 5 × 4 = 20 right angles.

ID	Question	Answer	Why (explain to student)
w2_s2	Set 3: Which of these shapes is a quadrilateral but is not a rectangle, a rhombus, or a square? [A shape with sides of 2, 3, 4, and 5 inches and no right angles / A shape with 4 sides of 3 inches each and 4 right angles / A shape with sides of 6, 2, 6, and 2 inches and 4 right angles / A shape with 3 sides of 4 inches each]	**A shape with sides of 2, 3, 4, and 5 inches and no right angles**	The shape with sides 2, 3, 4, and 5 inches has 4 straight sides, but no right angles and no equal sides. The 3-sided shape is a triangle.

TWO STEPS WITH ADDING AND SUBTRACTING (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
a1_s0	Set 1: The stand has 312 paper plates. Gemma uses 128 at lunch and 96 at dinner. Let p stand for the plates left. What is p?	**88** (also accepted: 88 plates, 88 paper plates, p = 88, p=88)	Step 1: 312 − 128 = 184. Step 2: 184 − 96 = 88. So p = 88 plates.
a2_s0	Set 1: Two steps! Hattie counts 145 people in line at noon. Then 68 more people get in line. After that, 120 people get their food and leave the line. Let n stand for the people still waiting. What is n?	**93** (also accepted: 93 people, n = 93, n=93)	Step 1: 145 + 68 = 213 people in line. Step 2: 213 − 120 = 93. So n = 93 people.
a1_s1	Set 2: A cooler holds 200 ears of corn. Hector grills 75 ears in the morning and 58 ears in the afternoon. Let c stand for the ears left in the cooler. What is c?	**67** (also accepted: 67 ears, 67 ears of corn, c = 67, c=67)	Step 1: 200 − 75 = 125. Step 2: 125 − 58 = 67. So c = 67 ears of corn.
a2_s1	Set 2: Find the mistake. Gemma solved 300 − 120 − 45 = n and got n = 225. What went wrong? [She added 45 instead of subtracting it. 300 − 120 = 180, and 180 − 45 = 135, so n = 135. / Nothing is wrong. n = 225. / She should multiply: 120 × 45. / She should start with 120 − 45 = 75, so n = 75.]	**She added 45 instead of subtracting it. 300 − 120 = 180, and 180 − 45 = 135, so n = 135.**	300 minus 120 is 180. 180 + 45 is 225, so Gemma added. The equation says subtract again: 180 − 45 = 135.
a1_s2	Set 3: The festival has 164 picnic tables in one field and 89 in another field. Then 47 tables are moved away to make room for the music stage. Let t stand for the tables left. What is t?	**206** (also accepted: 206 tables, 206 picnic tables, t = 206, t=206)	Step 1: 164 + 89 = 253 tables. Step 2: 253 − 47 = 206. So t = 206 tables.
a2_s2	Set 3: Two steps! Hattie has 500 napkins. She hands out 236 napkins before lunch and 147 after lunch. How many napkins does she have left?	**117** (also accepted: 117 napkins)	Step 1: 500 − 236 = 264. Step 2: 264 − 147 = 117 napkins.

TWO STEPS WITH MULTIPLYING AND DIVIDING (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
m1_s0	Set 1: Gemma sets 9 tables with 4 bottles of sauce on each table. Later she takes away 7 bottles that are empty. Let s stand for the bottles still on the tables. What is s?	**29** (also accepted: 29 bottles, s = 29, s=29)	Step 1: 9 × 4 = 36 bottles. Step 2: 36 − 7 = 29. So s = 29 bottles.
m2_s0	Set 1: Two steps! Mr. Hargrove shares 72 slices of brisket equally on 8 plates. Hattie eats 2 slices from one plate. How many slices are left on that plate?	**7** (also accepted: 7 slices)	Step 1: 72 ÷ 8 = 9 slices on each plate. Step 2: 9 − 2 = 7 slices.

ID	Question	Answer	Why (explain to student)
m1_s1	Set 2: Hector buys 5 packs of napkins with 10 napkins in each pack. He already had 23 napkins. Let n stand for all his napkins now. What is n?	**73** (also accepted: 73 napkins, n = 73, n=73)	Step 1: 5 × 10 = 50 napkins. Step 2: 50 + 23 = 73. So n = 73 napkins.
m2_s1	Set 2: Two steps! Mr. Hargrove smokes 48 sausages and splits them equally onto 6 trays. He sells 4 of the trays. How many sausages did he sell?	**32** (also accepted: 32 sausages)	Step 1: 48 ÷ 6 = 8 sausages on each tray. Step 2: 4 trays × 8 = 32 sausages.
m1_s2	Set 3: At the cooking contest, 7 teams each cook 6 racks of ribs. The judges taste 9 of the racks. How many racks are not tasted?	**33** (also accepted: 33 racks, 33 racks of ribs)	Step 1: 7 × 6 = 42 racks. Step 2: 42 − 9 = 33 racks.
m2_s2	Set 3: Which explanation is correct? Hattie has 30 ears of corn in 5 equal piles. She grills 2 of the piles. How many ears did she grill? [First divide: 30 ÷ 5 = 6 ears in each pile. Then multiply: 2 × 6 = 12 ears. / Subtract twice: 30 − 5 − 2 = 23 ears. / Divide by 2: 30 ÷ 2 = 15 ears. / Multiply: 5 × 2 = 10 ears.]	**First divide: 30 ÷ 5 = 6 ears in each pile. Then multiply: 2 × 6 = 12 ears.**	Sharing 30 into 5 equal piles gives 6 in each pile. 2 piles of 6 is 12 ears.

WRITING THE EQUATION (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
q1_s0	Set 1: Which equation matches the story? Gemma has 5 boxes with 9 cornbread squares in each box. She sells some squares and has 12 left. Let s be the number of squares she sold. [5 × 9 − s = 12 / 5 + 9 − s = 12 / 5 × 9 + 12 = s / s − 5 × 9 = 12]	**5 × 9 − s = 12**	She starts with 5 equal groups of 9, which is 5 × 9. She takes away the s she sold, and 12 are left.
q2_s0	Set 1: Solve the equation: 5 × 9 − s = 12. What is s?	**33** (also accepted: s = 33, s=33)	Step 1: 5 × 9 = 45. Step 2: 45 − s = 12, so s = 45 − 12 = 33.
q1_s1	Set 2: Solve the equation: k + 7 × 8 = 90. What is k?	**34** (also accepted: k = 34, k=34)	Step 1: 7 × 8 = 56. Step 2: k + 56 = 90, so k = 90 − 56 = 34.
q2_s1	Set 2: Which equation matches the story? Hector has 64 rolls. He puts the same number of rolls in each of 8 bags. Then he adds 3 more rolls to one bag. Let r be the number of rolls in that bag now. [64 ÷ 8 + 3 = r / 64 − 8 + 3 = r / 64 ÷ 3 + 8 = r / 64 × 8 + 3 = r]	**64 ÷ 8 + 3 = r**	Sharing 64 into 8 equal bags is 64 ÷ 8 = 8. Then 3 more makes 8 + 3 = 11 rolls.
q1_s2	Set 3: Solve the equation: 6 × 7 + g = 50. What is g?	**8** (also accepted: g = 8, g=8)	Step 1: 6 × 7 = 42. Step 2: 42 + g = 50, so g = 50 − 42 = 8.
q2_s2	Set 3: Find the mistake. Hattie solved 100 − 4 × 9 = n. She got n = 864, because 100 − 4 = 96 and 96 × 9 = 864. What went wrong? [She should multiply first: 4 × 9 = 36. Then 100 − 36 = 64, so n = 64. / Nothing is wrong. n = 864. / She should add: 100 + 4 + 9 = 113. / She should divide: 100 ÷ 4 = 25.]	**She should multiply first: 4 × 9 = 36. Then 100 − 36 = 64, so n = 64.**	When an equation has × and −, do the × part first. 4 × 9 = 36, and 100 − 36 = 64.

IS MY ANSWER REASONABLE? ESTIMATE BY ROUNDING (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: Round 467 to the nearest hundred.	**500**	467 is between 400 and 500. The tens digit is 6, which is 5 or more, so 467 rounds up to 500.
r2_s0	Set 1: Find the mistake. Hattie says 318 + 276 = 894. Use an estimate to find what went wrong. [Her answer is too far from the estimate 300 + 300 = 600. The real sum is 594. / Nothing is wrong. 894 is close to 600. / The estimate is 300 + 200 = 500, so the answer should be 500. / She should have subtracted: 318 − 276 = 42.]	**Her answer is too far from the estimate 300 + 300 = 600. The real sum is 594.**	318 rounds to 300 and 276 rounds to 300, so the sum should be near 600. The exact sum is 318 + 276 = 594.
r1_s1	Set 2: Estimate by rounding each number to the nearest ten: 48 + 33. What is your estimate?	**80**	48 rounds to 50 and 33 rounds to 30. 50 + 30 = 80.
r2_s1	Set 2: Two steps! The stand sold 186 pulled pork sandwiches and 223 brisket sandwiches. Round each number to the nearest hundred, then add to estimate the total.	**400** (also accepted: 400 sandwiches)	Step 1: 186 rounds to 200 and 223 rounds to 200. Step 2: 200 + 200 = 400. (The exact total, 409, is close.)
r1_s2	Set 3: Round 735 to the nearest ten.	**740**	735 is between 730 and 740. The ones digit is 5, so it rounds up to 740.
r2_s2	Set 3: Which explanation is correct? Hector solved 600 − 389 and got 211. Is his answer reasonable? [Yes. 389 rounds to 400, and 600 − 400 = 200. 211 is close to 200. / No. The estimate is 600 + 400 = 1,000. / No. 389 rounds to 300, so the answer should be 300. / Yes, because 211 is a three-digit number.]	**Yes. 389 rounds to 400, and 600 − 400 = 200. 211 is close to 200.**	Round 389 to 400. 600 − 400 = 200, and the exact answer 211 is close to 200, so it is reasonable.

PIT CHALLENGE: TWO-STEP PROBLEMS (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
c1_s0	Set 1: Two steps! Mr. Hargrove smokes 4 briskets on Friday and 5 briskets on Saturday. Each brisket makes 8 plates of food. How many plates is that?	**72** (also accepted: 72 plates)	Step 1: 4 + 5 = 9 briskets. Step 2: 9 × 8 = 72 plates.
c2_s0	Set 1: Two steps! The festival had 425 visitors in the morning. In the afternoon, 168 visitors went home and 239 new visitors arrived. How many visitors are there now?	**496** (also accepted: 496 visitors)	Step 1: 425 − 168 = 257. Step 2: 257 + 239 = 496 visitors. Check by rounding to the nearest ten: 430 − 170 + 240 = 500, which is close to 496.
c1_s1	Set 2: Two steps! Gemma packs 63 corn muffins into boxes, with 7 muffins in each box. She sells all but 2 of the boxes. How many boxes did she sell?	**7** (also accepted: 7 boxes)	Step 1: 63 ÷ 7 = 9 boxes. Step 2: 9 − 2 = 7 boxes sold.
c2_s1	Set 2: Two steps! A judge tastes 3 samples from each of 9 teams. Then she tastes 6 more samples from the kids' contest. How many samples does she taste in all?	**33** (also accepted: 33 samples)	Step 1: 9 × 3 = 27 samples. Step 2: 27 + 6 = 33 samples.

ID	Question	Answer	Why (explain to student)
c1_s2	Set 3: Two steps! Hector buys 6 bags of charcoal. Each bag costs $8. He pays with a $50 bill. How many dollars of change does he get?	**2** (also accepted: $2, 2 dollars, $2.00, 2 dollar, $ 2)	Step 1: 6 × $8 = $48. Step 2: $50 − $48 = $2 in change.
c2_s2	Set 3: Two steps! Hattie counted 274 bottles of sauce. She sold 96 bottles on Saturday and 105 bottles on Sunday. How many bottles are left?	**73** (also accepted: 73 bottles)	Step 1: 274 − 96 = 178. Step 2: 178 − 105 = 73 bottles. Check by rounding to the nearest ten: 270 − 100 − 110 = 60, which is close to 73.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match equation	**a number sentence with an equal sign**	2 points if matched on the first try.
match_s0_2	Set 1: Match unknown	**the number you need to find**	2 points if matched on the first try.
match_s0_3	Set 1: Match estimate	**a close answer found with easy numbers**	2 points if matched on the first try.
match_s0_4	Set 1: Match round	**change a number to the nearest ten or hundred**	2 points if matched on the first try.
match_s0_5	Set 1: Match reasonable	**makes sense and is close to the estimate**	2 points if matched on the first try.
match_s1_1	Set 2: Match 347 to the nearest hundred	**300**	2 points if matched on the first try.
match_s1_2	Set 2: Match 650 to the nearest hundred	**700**	2 points if matched on the first try.
match_s1_3	Set 2: Match 84 to the nearest ten	**80**	2 points if matched on the first try.
match_s1_4	Set 2: Match 85 to the nearest ten	**90**	2 points if matched on the first try.
match_s1_5	Set 2: Match 463 to the nearest ten	**460**	2 points if matched on the first try.
match_s2_1	Set 3: Match n + 25 = 60	**n = 35**	2 points if matched on the first try.
match_s2_2	Set 3: Match 4 × n = 28	**n = 7**	2 points if matched on the first try.
match_s2_3	Set 3: Match n − 18 = 30	**n = 48**	2 points if matched on the first try.
match_s2_4	Set 3: Match 54 ÷ n = 6	**n = 9**	2 points if matched on the first try.
match_s2_5	Set 3: Match 3 × 5 + n = 20	**n = 5**	2 points if matched on the first try.

FIX THE SENTENCE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
fs1_s0	Set 1: Fix the sentence: kansas city is famous for barbecue	**Kansas City is famous for barbecue.**	Kansas City is the name of a city, so both words get capital letters. End the telling sentence with a period.
fs2_s0	Set 1: Fix the sentence: we sold ribs brisket and corn	**We sold ribs, brisket, and corn.** (also accepted: We sold ribs, brisket and corn.)	Start with a capital W, put commas between the things in the list, and end with a period.
fs1_s1	Set 2: Fix the sentence: how many ribs are left	**How many ribs are left?**	Start with a capital H. This sentence asks something, so it ends with a question mark.

ID	Question	Answer	Why (explain to student)
fs2_s1	Set 2: Fix the sentence: gemma hector and hattie counted the plates	**Gemma, Hector, and Hattie counted the plates.** (also accepted: Gemma, Hector and Hattie counted the plates.)	Each name gets a capital letter, commas go between the names in the list, and the sentence ends with a period.
fs1_s2	Set 3: Fix the sentence: did mr. hargrove win a ribbon	**Did Mr. Hargrove win a ribbon?**	Start with a capital D. Mr. and Hargrove are part of a name, so they get capital letters. The sentence asks something, so it ends with a question mark.
fs2_s2	Set 3: Fix the sentence: hector and hattie is selling corn	**Hector and Hattie are selling corn.**	Start with a capital H, give both names capital letters, and end with a period. Two people means use are, not is.

BOSS BATTLE: THE DINNER RUSH (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb1_s0	Set 1: Two steps! Gemma made 180 cups of lemonade. In the morning she sold 4 trays with 9 cups on each tray. How many cups of lemonade are left?	**144** (also accepted: 144 cups)	Step 1: 4 × 9 = 36 cups sold. Step 2: 180 − 36 = 144 cups left.
bb2_s0	Set 1: Which explanation is correct? Hattie says 487 + 309 is about 800. [She is right. 487 rounds to 500 and 309 rounds to 300, and 500 + 300 = 800. / She is wrong. 487 rounds to 400, so it is about 700. / She is wrong. You cannot estimate a sum. / She is right, because 487 + 309 = 800 exactly.]	**She is right. 487 rounds to 500 and 309 rounds to 300, and 500 + 300 = 800.**	Rounding to the nearest hundred gives 500 + 300 = 800. The exact sum is 796, which is close.
bb1_s1	Set 2: Two steps! At the festival, 9 teams each bring 5 coolers. By evening, 12 of the coolers are empty. How many coolers are still full?	**33** (also accepted: 33 coolers)	Step 1: 9 × 5 = 45 coolers. Step 2: 45 − 12 = 33 full coolers.
bb2_s1	Set 2: Solve the equation: t − 6 × 7 = 18. What is t?	**60** (also accepted: t = 60, t=60)	Step 1: 6 × 7 = 42. Step 2: t − 42 = 18, so t = 18 + 42 = 60.
bb1_s2	Set 3: Two steps! The smoker holds 4 rows of chickens with 7 chickens in each row. Mr. Hargrove adds 9 more chickens on the top rack. How many chickens are in the smoker?	**37** (also accepted: 37 chickens)	Step 1: 4 × 7 = 28 chickens. Step 2: 28 + 9 = 37 chickens.
bb2_s2	Set 3: Find the mistake. Gemma estimated 712 − 288 by writing 700 − 200 = 500. What went wrong? [288 rounds to 300, not 200. The estimate should be 700 − 300 = 400. / Nothing is wrong. 288 rounds to 200. / 712 rounds to 800, so the estimate is 600. / You must add to estimate: 700 + 300 = 1,000.]	**288 rounds to 300, not 200. The estimate should be 700 − 300 = 400.**	The tens digit of 288 is 8, so it rounds up to 300. 700 − 300 = 400, and the exact answer 424 is close.

BOSS BATTLE: CLOSING TIME (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb3_s0	Set 1: Two steps! Mr. Hargrove sells barbecue plates for $9 each. He sells 7 plates, and then a happy customer gives him a $5 tip. How many dollars does he get in all?	**68** (also accepted: $68, 68 dollars, $68.00, 68 dollar, $ 68)	Step 1: 7 × $9 = $63. Step 2: $63 + $5 = $68.
bb4_s0	Set 1: Find the mistake. Hector says 234 + 158 − 100 = 492. Type the correct answer.	**292**	Step 1: 234 + 158 = 392. Step 2: 392 − 100 = 292. Hector added 100 instead of subtracting it.
bb3_s1	Set 2: Two steps! The festival map shows 6 rows of food stands with 8 stands in each row. 9 of the stands sell barbecue and the rest sell other foods. How many stands sell other foods?	**39** (also accepted: 39 stands, 39 food stands)	Step 1: 6 × 8 = 48 stands. Step 2: 48 − 9 = 39 stands.
bb4_s1	Set 2: Find the mistake. Gemma says 5 × 6 + 4 = 50, because 6 + 4 = 10 and 5 × 10 = 50. Type the correct answer.	**34**	Do the × part first: 5 × 6 = 30. Then 30 + 4 = 34.
bb3_s2	Set 3: Two steps! 342 people came to the festival on Saturday and 279 came on Sunday. 150 of all those people were cooks and judges. How many were visitors?	**471** (also accepted: 471 visitors, 471 people)	Step 1: 342 + 279 = 621 people. Step 2: 621 − 150 = 471 visitors. Check by rounding to the nearest ten: 340 + 280 − 150 = 470, which is close to 471.
bb4_s2	Set 3: Find the mistake. Hattie solved 40 − 24 ÷ 4 = n. She said n = 4, because 40 − 24 = 16 and 16 ÷ 4 = 4. Type the correct value of n.	**34** (also accepted: n = 34, n=34)	Do the ÷ part first: 24 ÷ 4 = 6. Then 40 − 6 = 34, so n = 34.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade3/prepost/KCE_Math_PrePost_G3W24.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: napkin with 4 equal sides and 4 right angles	Most exact name: square. Also fits: rectangle, rhombus or quadrilateral.
2	40 ribs; sells 12, then 9 more	40 − 12 = 28, and 28 − 9 = 19 ribs left.
3	5 corn muffins in each of 4 baskets	4 × 5 = 20 muffins.
4	Round 68 to the nearest ten	70
5	15 + n = 22: what does n stand for?	7 (22 − 15 = 7).

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	400 plates; 165 at lunch and 128 at dinner: equation and answer	Example: 400 − 165 − 128 = p. Step 1: 400 − 165 = 235. Step 2: 235 − 128 = 107. p = 107 plates. Accept any letter and any correct equation.
2	54 burnt ends shared on 6 plates; eats 3 from one plate	Step 1: 54 ÷ 6 = 9 on each plate. Step 2: 9 − 3 = 6 burnt ends left on that plate.

#	Prompt	Model answer / grading note
3	Gemma says 8 × 5 − 4 = 8	She subtracted first. Do the × part first: 8 × 5 = 40, and 40 − 4 = 36.
4	Estimate 326 + 471 to the nearest hundred; exact sum	Estimate: 300 + 500 = 800. Exact: 326 + 471 = 797. It is reasonable because 797 is very close to 800.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Grady has 7 packs of 8 cups and uses 39	Example: 7 × 8 − 39 = c. Step 1: 7 × 8 = 56. Step 2: 56 − 39 = 17. c = 17 cups left.